Package 'golemenviro'

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Title Real-time visualization of Pimoroni Enviro sensor data from Prometheus

Version 0.0.0.9000

Description A Shiny application, built as a golem package, that displays the readings of Pimoroni Enviro and Enviro+ sensors exported to Prometheus: gauges with the latest readings, and a time series chart per sensor.

License GPL-3

URL https://sedzinfo.github.io/golemenviro/, https://github.com/sedzinfo/golemenviro

BugReports https://github.com/sedzinfo/golemenviro/issues

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`get_data` *Get Data from Prometheus*

Description

Downloads the readings of all the sensors of a Pimoroni Enviro or Enviro+ from Prometheus, at about 1000 points in time between `start_time` and `end_time`.

Usage

```
get_data(
  start_time = format(Sys.time() - 60 * 60 * 24 * 7, "%Y-%m-%dT%H:%M:%SZ"),
  end_time = format(Sys.time(), "%Y-%m-%dT%H:%M:%SZ"),
  urlq
)
```

Arguments

`start_time`	start time in this format: "2025-01-31T11:48:08Z"
`end_time`	end time in this format: "2025-02-07T11:48:33Z"
`urlq`	Prometheus url

Value

A data frame with the Date and the Timestamp of the readings and one column per sensor, or `NULL`, with a message, when the server cannot be reached or sends no data. Sensors whose readings never change are left out, as the device does not have them.

Examples

```
prometheus_url <- c(
  "http://pip1.crabdance.com:1507/api/v1/query_range",
  "http://pip1.crabdance.com:1505/api/v1/query_range"
)
start_time <- format(Sys.time() - 60 * 60 * 24 * 7, "%Y-%m-%dT%H:%M:%SZ")
end_time <- format(Sys.time(), "%Y-%m-%dT%H:%M:%SZ")
filtered_df <- get_data(start_time = start_time, end_time = end_time, urlq = prometheus_url[2])
head(filtered_df)
filtered_df <- get_data(start_time = start_time, end_time = end_time, urlq = prometheus_url[1])
head(filtered_df)
```

pimoroni *Run the Pimoroni Enviro app*

Description

Starts a Shiny app with the readings of Pimoroni Enviro and Enviro+ sensors exported to Prometheus: gauges with the latest readings in the Current tab, and a chart per sensor in the Time Series tab.

Usage

```
pimoroni(
  prometheus_url,
  timezone = "Europe/Bucharest",
  onStart = NULL,
  options = list(),
  enableBookmarking = NULL,
  uiPattern = "/",
  ...
)
```

Arguments

prometheus_url	Character vector of Prometheus query_range urls, one per device, such as "http://pip1.crabdance.com:150
timezone	The time zone the readings are shown in, while the app runs.
onStart	See shiny::shinyApp().
options	See shiny::shinyApp().
enableBookmarking	See shiny::shinyApp().
uiPattern	See shiny::shinyApp().
...	Further options passed to golem::with_golem_options().

Value

A shiny::shinyApp() object. Printing it, or calling pimoroni() at the console, opens the app in the browser.

Examples

```
if (interactive()) {
  pimoroni(prometheus_url = c(
    "http://pip1.crabdance.com:1507/api/v1/query_range",
    "http://pip1.crabdance.com:1505/api/v1/query_range"
  ))
}
```

plot_gauge *Plot Gauge*

Description

Plots the last reading of one sensor as a gauge, with the ideal range in green and a short note on what the gas and particle sensors measure.

Usage

```
plot_gauge(data, variable = "Temperature")
```

Arguments

data	data frame from get_data()
variable	one of "Temperature" "Humidity" "Pressure" "Lux" "Proximity" "NH3" "Reducing" "Oxidising" "PM1" "PM2.5" "PM10"

Value

A plotly object.

Examples

```
prometheus_url <- c(
  "http://pip1.crabdance.com:1507/api/v1/query_range",
  "http://pip1.crabdance.com:1505/api/v1/query_range"
)
filtered_df <- get_data(urlq = prometheus_url[1])
plot_gauge(data = filtered_df, variable = "Temperature")
plot_gauge(data = filtered_df, variable = "Humidity")
plot_gauge(data = filtered_df, variable = "PM2.5")
```

plot_time_series *Plot Time Series*

Description

Plots the readings of one sensor, with their moving average, their mean, and a summary of the last reading, mean, minimum, maximum, standard deviation, range and dates on the right.

Usage

```
plot_time_series(data, variable = "Temperature", scale = "", k = 30)
```

Arguments

data	data frame from get_data()
variable	one of "Temperature" "Humidity" "Pressure" "Lux" "Proximity" "NH3" "Reducing" "Oxidising" "PM1" "PM2.5" "PM10"
scale	title of the y axis
k	integer width of the rolling window. Must be odd for rollmedian

Value

A plotly object.

Examples

```
prometheus_url <- c(
  "http://pip1.crabdance.com:1507/api/v1/query_range",
  "http://pip1.crabdance.com:1505/api/v1/query_range"
)
filtered_df <- get_data(urlq = prometheus_url[1])
plot_time_series(data = filtered_df, variable = "Temperature")
plot_time_series(data = filtered_df, variable = "Humidity")
plot_time_series(data = filtered_df, variable = "Pressure")
plot_time_series(data = filtered_df, variable = "PM2.5")
```

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